Free entropy dimension of free group factors: the derivation chain, and the invariance step that is missing

Multi-agent collaborative research group

October 2026

摘要

Let F_k be the free group of rank k and $W^*(F_k) = L(F_k)$ the corresponding free group factor; the question is whether $W^*(F_k) \cong W^*(F_m)$ can hold for $k \neq m$. We do not answer it. We assemble the known values of the four free entropy dimensions into a chain a reader can check line by line: an inequality chain, the ℓ^2-Betti formula of Mineyev and Shlyakhtenko, its evaluation on free groups, invariance of the non-microstates dimensions under change of generating tuple, and the resulting value k on every algebraic generating tuple of $\mathbb{C}F_k$. Every step is attributed to a definition, to an earlier result of this paper, or to a named published theorem; the two external inputs are stated as such at the step where they are used. We then show by an elementary argument that turning this value into a statement about factors requires exactly one missing ingredient, the invariance of the microstates dimension δ_0 under von Neumann algebra isomorphism; it is stated as a conjecture and is not proved here. An elementary embedding lemma shows that no coarse obstruction separates the two factors. No estimator of δ_0 with controlled error was available to us, and we state what such an estimator would have to satisfy; the computations we did run are collected in Appendix A.

1 Introduction

Let $k \geq 2$, let $F_k = \langle u_1, \ldots, u_k \rangle$ be the free group of rank k, and let $W^*(F_k) = L(F_k)$ be the von Neumann algebra generated by the left regular representation of F_k. $W^*(F_k)$ is a II_1 factor, the free group factor of rank k. The question studied here is: *is $W^*(F_k)$ isomorphic to $W^*(F_m)$ for $k \neq m$?* Neither direction is settled; this paper does not settle either.

Why the question matters is not internal to free groups. $W^*(F_k)$ is the most transparent family of II_1 factors, and *rank* is the only parameter attached to it. If rank is not recoverable from the isomorphism class, then any classification of II_1 factors by invariants must be formulated without rank as a primitive, and the free group factors become the standard test case for that reformulation. If rank *is* recoverable, then the invariant which recovers it must satisfy two demands simultaneously: it must be invariant under von Neumann algebra isomorphism, and it must be computable from

the data of a generating tuple. No currently known invariant is proved to satisfy both; the free entropy dimensions satisfy the second demand on group algebras and the first only conjecturally.

What is known. Dykema constructed the interpolated free group factors $L(F_t)$, $1 < t \leq \infty$, a continuous family whose integer points are exactly the free group factors, so that the integer question sits inside a real-parameter one [Dy94]. Popa and Shlyakhtenko showed that every member of that family is isomorphic to a genuine group factor [PS20], and Houdayer produced strongly solid group factors outside the family [Hou09]; thus membership in the family is a nontrivial property. On the free entropy side, Voiculescu introduced the microstates free entropy χ and the free entropy dimensions δ, δ_0 [Voi94], and used them to show that some generators of property (T) factors satisfy $\delta \leq 1$ [Voi99]. Mineyev and Shlyakhtenko computed the non-microstates dimensions of group algebras: for a finitely generated group G and self-adjoint generators of $\mathbb{C}G$, $\delta_* = \delta^\star = \beta_1(G) - \beta_0(G) + 1$ [MS05]. Shlyakhtenko's short survey note records that on the group algebra of F_k all four dimensions agree and equal k [Sh06]. The ℓ^2-Betti numbers used in the formula are Atiyah's; their relation to the L^2-Euler characteristic is in [At76].

The contribution of this paper is expository and elementary, and each item points to a numbered result below. (i) Lemma 3.1 records the inequality chain $\delta_0 \leq \delta \leq \delta_* \leq \delta^\star$ step by step, naming the source of each comparison. (ii) Lemma 3.2 shows that the self-adjoint generating tuples required by [MS05] exist for every k, and exhibits them. (iii) Lemma 3.4 computes $\beta_0(F_k) = 0$ and $\beta_1(F_k) = k - 1$ from the L^2-Euler characteristic rather than quoting them; Proposition 3.5 then evaluates the Mineyev–Shlyakhtenko formula to k. (iv) Lemma 3.6 isolates why the non-microstates dimensions do not depend on the chosen generating tuple, and Proposition 3.7 combines it with the chain to obtain $\delta_0 = \delta = \delta_* = \delta^\star = k$ on *every* algebraic generating tuple of $\mathbb{C}F_k$. (v) Lemma 3.9 and Proposition 3.10 explain why this value does not yet decide the question: $W^*(F_k)$ embeds in $W^*(F_m)$, so no coarse obstruction exists, and separating the factors through δ_0 presupposes that δ_0 is an isomorphism invariant of the factor, which is Conjecture 3.11 and is not proved. (vi) Open problem 4.1 states what a computable estimator of δ_0 would have to satisfy; we did not have one, so none of our numerical work bears on the question, and it is confined to Appendix A.

The paper is organised by the argument, not by the order in which the material was produced. Section 2 fixes definitions and notation and formulates the question precisely. Section 3 contains the results and their proofs and is the body of the paper. Section 4 discusses the boundary of what is proved, the conjectures, and the open problems. References follow the main text, and Appendix A records the computations, with algorithms, parameter ranges and numerical readings.

2 Definitions and notation

Throughout, $k, m \geq 2$ are integers, $F_k = \langle u_1, \ldots, u_k \rangle$ is the free group of rank k with identity e, and $S = \{u_1^{\pm 1}, \ldots, u_k^{\pm 1}\}$ is its standard generating set. For $w \in F_k$, $|w|$ is the number of letters

of the reduced word representing w, and $B_r = \{w \in F_k : |w| \le r\}$.

Definition 2.1 (Group algebra and trace)**.** $\mathbb{C}F_k$ is the set of formal finite sums $\sum_{g\in F_k} \alpha_g g$ with $\alpha_g \in \mathbb{C}$ and only finitely many $\alpha_g \neq 0$, with multiplication extending the group law, involution $(\sum_g \alpha_g g)^* = \sum_g \overline{\alpha_g} g^{-1}$, and unit e. The functional $\tau(\sum_g \alpha_g g) = \alpha_e$ is a faithful tracial state: $\tau(1) = 1$, $\tau(xy) = \tau(yx)$, and $\tau(x^*x) = 0$ implies $x = 0$.

Definition 2.2 (Free group factor)**.** Let $\ell^2(F_k)$ have orthonormal basis $(\xi_g)_{g\in F_k}$ and let $\lambda(g)\xi_h = \xi_{gh}$ be the left regular representation. Put $W^*(F_k) = L(F_k) = \{\lambda(g) : g \in F_k\}''$. It is a II_1 factor with trace $\tau(x) = \langle x\xi_e, \xi_e\rangle$, and λ extends to an injective $*$-homomorphism of $\mathbb{C}F_k$ with weakly dense image; we write $u_j = \lambda(u_j)$ and identify $\mathbb{C}F_k$ with its image.

Definition 2.3 (Microstates)**.** For an n-tuple $X = (X_1, \dots, X_n)$ of self-adjoint elements of a tracial $*$-probability space (M, τ), for $N \ge 1$, $R > 0$, $\varepsilon > 0$ and $s \ge 1$, let $\Gamma_R(X; N, \varepsilon, s)$ be the set of tuples $(A_1, \dots, A_n)$ of self-adjoint $N \times N$ matrices such that $\|A_j\| \le R$ for all j and $|\operatorname{tr}(A_{j_1} \cdots A_{j_s}) - \tau(X_{j_1} \cdots X_{j_s})| < \varepsilon$ for all words of length at most s. Vol denotes Lebesgue measure on $(M_N(\mathbb{C})^{sa})^n$.

Definition 2.4 (Free entropy dimensions)**.** $\delta(X)$ and $\delta_0(X)$ are Voiculescu's microstates free entropy dimensions [Voi94]: the exponential growth rate in N^2 of the Lebesgue measure of the ε-neighbourhood of $\Gamma_R(X; N, \varepsilon, s)$, normalised so that a ball of radius ε contributes zero, taken with a limit superior in N followed by $\varepsilon \to 0$ for δ, and with the additional vanishing condition on the cutoff for δ_0. $\delta_*(X)$ and $\delta^\star(X)$ are the non-microstates free entropy dimensions, defined through the free entropy of finite-dimensional approximations and its modification by a cut-off function [Voi94]. We do not reproduce the normalising constants here; the three properties of these four quantities that the proofs below use are the inequality chain of Lemma 3.1, the dependence of $\delta_*, \delta^\star$ on the algebra only through the group (Theorem 3.3), and the algebraic invariance of δ_0 stated at step (2) of Proposition 3.7, each cited where used.

Definition 2.5 (ℓ^2-Betti numbers, Euler characteristic)**.** For a discrete group G with a free resolution of $\mathbb{C}G$-modules of finite type, $\beta_i(G)$ is the i-th ℓ^2-Betti number, i.e. the Murray–von Neumann dimension of the i-th L^2-homology with respect to the group von Neumann algebra [At76]. When G has a finite classifying space X with $\chi(X) = \sum_i (-1)^i \dim_{\mathbb{C}} H_i(X; \mathbb{C})$, one writes $\chi(G)$ for that integer.

Definition 2.6 (Isomorphism question)**.** $W^*(F_k)$ and $W^*(F_m)$ are isomorphic if there is a bijective $*$-homomorphism $\theta : W^*(F_k) \to W^*(F_m)$; such a θ automatically preserves the trace, since traces on II_1 factors are unique.

Remark 2.7. By Definition 2.4 the dimensions are attached to tuples, not to algebras. That distinction is the whole difficulty of this paper: the statement we can prove is about tuples in $\mathbb{C}F_k$, the

statement needed for Question 2.6 is about isomorphism classes of factors, and the passage from the first to the second is Conjecture 3.11.

3 Main results

Every result below is stated with its hypotheses and proved in three numbered steps. A step is a consequence of a definition from Section 2, of an earlier result of this paper, or of a published theorem, which is named at that step.

Lemma 3.1 (Inequality chain)**.** *Let $X = (X_1, \ldots, X_n)$ be a self-adjoint n-tuple in a tracial $*$-probability space. Then*

$$\delta_0(X) \leq \delta(X) \leq \delta_*(X) \leq \delta^\star(X).$$

证明. (1) All four quantities in the display are defined, and the tuple is self-adjoint, so Voiculescu's definitions [Voi94] apply to it; this is the hypothesis check and it uses Definition 2.4 only. (2) The first two comparisons, $\delta_0 \leq \delta \leq \delta_*$, compare families of microstate spaces (respectively a microstate quantity with a non-microstates one); they are recorded with their references in [Sh06, Lemma 6], and we use them there as cited inputs, since reproving them would be a separate paper. (3) The third comparison $\delta_* \leq \delta^\star$ is part of the same chain in [Sh06, Lemma 6]; concatenating (2) and (3) gives the display. □

Lemma 3.2 (Self-adjoint generators)**.** *For every $k \geq 1$ the elements $a_j = u_j + u_j^{-1}$, $b_j = i(u_j - u_j^{-1})$, $1 \leq j \leq k$, are self-adjoint and generate $\mathbb{C}F_k$ as an algebra.*

证明. (1) $a_j^* = u_j^{-1} + u_j = a_j$ and $b_j^* = -i(u_j^{-1} - u_j) = i(u_j - u_j^{-1}) = b_j$ by Definition 2.1, so both families are self-adjoint. (2) From the two displayed formulas, $u_j = (a_j - ib_j)/2$ and $u_j^{-1} = (a_j + ib_j)/2$. (3) Hence every element of $S^{\pm 1}$, and therefore every word in F_k, lies in the algebra generated by $\{a_j, b_j\}$; since these words span $\mathbb{C}F_k$ by Definition 2.1, that algebra is $\mathbb{C}F_k$. □

Theorem 3.3 (Mineyev–Shlyakhtenko [MS05])**.** *Let G be a finitely generated discrete group and let $X_1, \ldots, X_n \in \mathbb{C}G$ be self-adjoint elements which generate $\mathbb{C}G$ as an algebra. Then*

$$\delta_*(X_1, \ldots, X_n) = \delta^\star(X_1, \ldots, X_n) = \beta_1(G) - \beta_0(G) + 1.$$

证明. (1) The quantity on the right is defined from the L^2-homology of G (Definition 2.5) and depends on G alone, not on the tuple. (2) The main result of [MS05] is exactly the displayed equality for self-adjoint generating tuples; we use it as an external input and do not reproduce its proof, which proceeds through an approximation of the non-microstates free entropy by finite-dimensional quotients. (3) Combining (1) and (2): both non-microstates dimensions take a value determined by G, so in particular they coincide with each other and are unchanged when the generating tuple is replaced. □

Lemma 3.4 (ℓ^2-Betti numbers of F_k)**.** *$\beta_0(F_k) = 0$ and $\beta_1(F_k) = k - 1$ for $k \geq 2$; more generally $\beta_i(F_k) = 0$ for $i \geq 2$.*

证明. (1) The wedge X of k circles is a classifying space for F_k, with $\chi(X) = 1 - k$; by Definition 2.5 this integer equals $\sum_i (-1)^i \beta_i(F_k)$ with the ℓ^2-Betti numbers of F_k, the equality of the L^2-Euler characteristic with the ordinary one being valid for groups with a finite classifying space [At76]. (2) F_k is infinite and non-amenable, so its Murray–von Neumann dimension in degree 0 vanishes, $\beta_0(F_k) = 0$; and X is one-dimensional, whence $\beta_i(F_k) = 0$ for $i \geq 2$. (3) Substituting (2) into (1) gives $1 - k = -\beta_1(F_k)$, i.e. $\beta_1(F_k) = k - 1$. □

Proposition 3.5 (The non-microstates dimensions of $\mathbb{C}F_k$)**.** *For every $k \geq 2$ and every self-adjoint generating n-tuple X of $\mathbb{C}F_k$,*

$$\delta_*(X) = \delta^\star(X) = k.$$

证明. (1) By Lemma 3.2 self-adjoint generating tuples of $\mathbb{C}F_k$ exist, so Theorem 3.3 applies with $G = F_k$. (2) By Lemma 3.4, $\beta_1(F_k) - \beta_0(F_k) + 1 = (k - 1) - 0 + 1 = k$. (3) Substituting (2) into the formula of Theorem 3.3 gives the display. □

Lemma 3.6 (Independence of the generating tuple)**.** *Let X and Y be two self-adjoint generating tuples of $\mathbb{C}F_k$. Then $\delta_*(X) = \delta_*(Y)$ and $\delta^\star(X) = \delta^\star(Y)$.*

证明. (1) Both tuples are self-adjoint and generate $\mathbb{C}F_k$, so Theorem 3.3 applies to each of them with the same group $G = F_k$. (2) By that theorem, each of the four numbers $\delta_*(X), \delta^\star(X), \delta_*(Y), \delta^\star(Y)$ equals the single number $\beta_1(F_k) - \beta_0(F_k) + 1$, which by step (1) of the proof of Theorem 3.3 involves only G. (3) Hence the four numbers coincide, which is the assertion. □

Proposition 3.7 (All four dimensions on $\mathbb{C}F_k$)**.** *Let $X = (X_1, \ldots, X_n)$ be any algebraic generating tuple of $\mathbb{C}F_k$ (self-adjointness not assumed). Then*

$$\delta_0(X) = \delta(X) = \delta_*(X) = \delta^\star(X) = k.$$

证明. (1) If X is self-adjoint, Lemma 3.1 gives $\delta_0(X) \leq \delta(X) \leq \delta_*(X) \leq \delta^\star(X)$ and Proposition 3.5 gives $\delta^\star(X) = k$, so all four are at most k; for a general tuple one first replaces it by the self-adjoint tuple of Lemma 3.2 built from the same generators, which changes none of the four quantities by the algebraic invariance used in step (3) below. (2) For the standard tuple $u_1, \ldots, u_k$, Shlyakhtenko records $\delta_0(u_1, \ldots, u_k) = k$ [Sh06, Lemma 6]; this is the one place where the value of δ_0 is taken from the literature rather than derived here. (3) δ_0 is an algebraic invariant: its value is unchanged when the generating tuple of the same algebra $\mathbb{C}F_k$ is replaced [Sh06, Lemma 6]. Combining with (2) gives $\delta_0(X) = k$, and with the bound of (1) this forces $\delta_0(X) = \delta(X) = \delta_*(X) = \delta^\star(X) = k$. □

Remark 3.8. Proposition 3.7 is a statement about tuples in the algebra $\mathbb{C}F_k$. It says that the four dimensions cannot tell two generating tuples of $\mathbb{C}F_k$ apart, and hence cannot, by themselves, tell $\mathbb{C}F_k$ from $\mathbb{C}F_m$ unless one already knows that some one of them passes to the weak closure in an isomorphism-invariant way.

Lemma 3.9 (Embedding of the smaller factor)**.** *For $k < m$ the map $\lambda(g) \mapsto \lambda(g)$, $g \in F_k$, where F_k is viewed as the subgroup of F_m generated by $u_1, \dots, u_k$, extends to an injective trace-preserving $*$-homomorphism $W^*(F_k) \to W^*(F_m)$.*

证明. (1) The subgroup generated by $u_1, \dots, u_k$ inside F_m is isomorphic to F_k, and $\ell^2(F_k)$ is the closed span of $\{\xi_g : g \in F_k\}$, a subspace of $\ell^2(F_m)$. (2) For $g \in F_k$ the operator $\lambda(g)$ on $\ell^2(F_m)$ preserves that subspace by Definition 2.2, and its restriction is the left regular representation of F_k; hence $\lambda(F_k)''$ is a von Neumann subalgebra of $\lambda(F_m)''$. (3) The inclusion is trace-preserving because both traces are $x \mapsto \langle x\xi_e, \xi_e\rangle$ with the same unit e, and injective because a trace-preserving $*$-homomorphism between finite factors is injective. □

Proposition 3.10 (What invariance of δ_0 would give)**.** *Assume that δ_0, computed on generating tuples, depends only on the isomorphism class of the von Neumann algebra generated. Then $W^*(F_k) \ncong W^*(F_m)$ for all $k \neq m$.*

证明. (1) Suppose $\theta : W^*(F_k) \to W^*(F_m)$ is an isomorphism, and let X be a generating tuple of $\mathbb{C}F_k$ with $\delta_0(X) = k$, which exists by Proposition 3.7; then $\theta(X)$ is a generating tuple of $W^*(F_m)$ and, by the assumed invariance, $\delta_0(\theta(X)) = \delta_0(X) = k$. (2) Applying Proposition 3.7 on the side of m to a generating tuple of $\mathbb{C}F_m$ obtained from $\theta(X)$ gives $\delta_0 = k$; applying it to the standard tuple of $\mathbb{C}F_m$ gives $\delta_0 = m$. (3) Steps (1) and (2) force $k = m$, contradicting $k \neq m$; hence no isomorphism exists. □

Conjecture 3.11 (Invariance of the microstates dimension)**.** *Let M and N be II_1 factors and $\theta : M \to N$ an isomorphism. Then for every finite generating tuple X of M, the value $\delta_0(X)$ equals $\delta_0(\theta(X))$ computed in N. In particular $\delta_0(W^*(F_k)) := k$ is well defined on the isomorphism class.*

Remark 3.12. What is proved in Proposition 3.7 is invariance of δ_0 under change of generating tuple *within one algebra*; Conjecture 3.11 asks for invariance under change of algebra. The gap between the two statements is not technical: the known proofs of tuple-invariance use that the generators lie in the same $\mathbb{C}F_k$ and compare finite tuples inside it, and no argument in the literature we could consult transports the microstate spaces across an abstract isomorphism of the weak closures.

4 Discussion and open problems

What is proved and what is not. The results of Section 3 are of two kinds. Lemma 3.4, Lemma 3.2 and Lemma 3.9 are elementary and proved in full. Theorem 3.3 is an external theorem, used at

the two places where its hypotheses are checked; Proposition 3.5, Lemma 3.6 and Proposition 3.7 are derived from it. Nothing in this paper decides whether $W^*(F_k) \cong W^*(F_m)$; the only statement about that question is the conditional Proposition 3.10, whose hypothesis is Conjecture 3.11.

Why the invariance step is the crux. By Proposition 3.7 the number k is a function of the tuple, not of the factor; by Lemma 3.9 the smaller factor sits inside the larger one with the same trace, so nothing about size or generation separates them. A separation argument must therefore produce a quantity defined on isomorphism classes. The free entropy dimensions are the natural candidates precisely because Voiculescu's property (T) computation [Voi99] shows they can be forced below the value expected for a free group factor; but that computation is again a statement about chosen generators of a specific algebra, and it does not close the invariance gap either.

A second boundary. Proposition 3.10 shows that Conjecture 3.11 would suffice; it does not show it is necessary, and we make no claim that every route to the question passes through δ_0. The interpolated family of [Dy94], the representation theorem of [PS20] and the exterior examples of [Hou09] all constrain which factors can be free group factors; none of them, as far as the sources at our disposal say, distinguishes ranks.

Open problem 4.1 (A qualified estimator of δ_0)**.** Find a computable procedure which, given k and a level N, outputs an estimate $\hat{\delta}_N$ of δ_0 for the standard generators of $\mathbb{C}F_k$, together with an error bound, and which satisfies: (i) convergence, with a relative standard error that tends to zero as $N \to \infty$ at a rate bounded a priori; (ii) stability in N, i.e. the relative change of $\hat{\delta}_N$ between successive levels below a fixed threshold, so that the sequence does not diverge with N; (iii) invariance under the normalisation chosen for the matrix model, i.e. agreement of the estimates up to a fixed relative tolerance when the same tuple is represented at different normalisation scales; (iv) feasibility on the whole parameter range, i.e. no level at which the estimate is vacuous. The missing step is the second half of (i) together with (iii): no proof or numerical evidence at our disposal controls the error of any estimator we could implement, and without error control the readings cannot distinguish the value k from an artefact of the truncation. Consequently no numerical result in this paper bears on Conjecture 3.11 or on the isomorphism question.

Open problem 4.2 (Invariance, quantitatively)**.** Even assuming Conjecture 3.11, it is open to bound how fast δ_0 of an approximating tuple approaches the value of the factor, i.e. to convert an invariance statement into a criterion which can be checked at a finite level. Such a bound would turn Open problem 4.1 into a test of the conjecture.

Numerical evidence. Numerical evidence is collected in Appendix A; nothing in it is used in the proofs of Section 3.

参考文献

[Dy94] K. J. Dykema, *Interpolated free group factors*, Pacific J. Math. **163** (1994), 123–135.

[PS20] S. Popa and D. Shlyakhtenko, *Representing interpolated free group factors as group factors*, Groups Geom. Dyn. **14** (2020).

[Hou09] C. Houdayer, *Strongly solid group factors which are not interpolated free group factors*, Math. Ann. **346** (2010), 969–989.

[Voi94] D. V. Voiculescu, *The analogues of entropy and of Fisher's information measure in free probability theory II*, Invent. Math. **118** (1994), 411–440.

[Voi99] D. V. Voiculescu, *Free entropy dimension* ≤ 1 *for some generators of property T factors of type* II_1, J. Reine Angew. Math. **514** (1999), 113–118.

[MS05] I. Mineyev and D. Shlyakhtenko, *Non-microstates free entropy dimension for groups*, Geom. Funct. Anal. **15** (2005), 476–490.

[Sh06] D. Shlyakhtenko, *Remarks on free entropy dimension*, in: Operator Algebras and their Applications, Fields Inst. Commun., Springer, 2006, 165–173.

[At76] M. F. Atiyah, *Elliptic operators and compact groups*, Lecture Notes in Math. **401**, Springer, 1974.

A Computational evidence

This appendix records the computations we ran. All readings below are outputs of scripts; none of them is used in the proofs of Section 3, and, as stated in Open problem 4.1, none of them decides anything about the isomorphism question.

A.0 Index of the statements the main text depends on. The proofs of Section 3 use exactly two statements, both taken from the literature and both registered with origin `external_lit`; each was checked verbatim against a local copy of the source (`quote_verified=true`).

register	statement	page	verbatim check
`stmt:000000`	Shlyakhtenko 2006, Lemma 6: $\delta_0 = \delta = \delta_* = \delta^\star = k$ on any generating tuple of $\mathbb{C}F_k$	7	passed
`stmt:000001`	Mineyev–Shlyakhtenko 2005, main theorem: $\delta^* = \delta^\star = \beta_1(G) - \beta_0(G) + 1$	2	passed

A.0.1 What the main text depends on. None of the results of Section 3 depends on a number produced by a script: every hypothesis there is either a definition or one of the two literature statements indexed in A.0. The computations listed below are exploratory only and carry no weight.

A.0.2 Exploratory computations (not load-bearing). All rows have method `numeric`; domain, exit code and script path are copied from the corresponding ledger row.

register	claim key	domain (ledger)	result	exit, script
`exp:000000`	fk-truncation-moments-k-invariance	$k \in \{2,3,4,8\}$, $n \in \{8,16,32,64\}$, samples≤ 40, seed 20261018	refutes	0, `code/fk_truncati`
`exp:000001`	delta_N_Xk_conj000000	seed 102008	refutes	0, `code/f_conj00000`
`exp:000002`	fk-truncation-kdep-scaling-check	$n \leq 128$, seed 20261019	inconclusive	0, `code/fk_truncati`
`exp:000003`	delta_N_Xk_conj000000	$n \leq 128$, seed 20261019	supports	0, `code/f_conj00000`
`exp:000004`	fk-truncation-corrected-kdep-diagnostic	$n \leq 128$	inconclusive	0, `code/fk_truncati`
`exp:000005`	fk-truncation-corrected-kdep-diagnostic	$n \leq 128$, seed 20261020	refutes	0, `code/fk_truncati`

A.0.3 The remaining exploratory rows. These five rows are exploratory as well; their `params` field is empty, so the parameter range is fixed by constants inside the script and recorded in the `observations` field of the ledger row.

register	claim_key	result	exit	script
`exp:000006`	conj000000.estimator_infeasible	inconclusive	0	`scripts/conj000000_infeasible_reco`
`exp:000007`	conj000000.estimator_infeasible	inconclusive	1	`scripts/conj000000_infeasible_reco`
`exp:000008`	conj000000.estimator_infeasible	inconclusive	0	`scripts/conj000000_infeasible_reco`
`exp:000009`	fk-microstate-entropy-dimension-k-separation	inconclusive	0	`code/conj000000_verdict_run.py`
`exp:000010`	fk-microstate-entropy-dimension-k-separation	inconclusive	0	`code/conj000000_verdict_run.py`

A.1 Literature cited in the main text. Each reference of the bibliography is matched below to its ledger row. "Local copy" means a plain-text full text is present in `library/`; "verbatim" refers to the `quote_verified` field, which is carried by the `statements` rows and not by the `literature` rows.

reference	ledger row	local copy	verbatim
[Dy94]	`lit:000000`	no	not checked
[PS20]	`lit:000001`	no	not checked
[Hou09]	`lit:000002`	no	not checked
[Voi99]	`lit:000003`	no	not checked
[Voi94]	no ledger row	no	not checked
[MS05]	`lit:000009`	yes	`true` (`stmt:000001`)
[Sh06]	`lit:000004`	yes	`true` (`stmt:000000`)
[At76]	no ledger row	no	not checked

A.2 How to re-run. Every command runs from the project root. The expected reading is the exit code together with the `result` field of the same ledger row; where the row records a headline number it is quoted as-is, and no expectation beyond the ledger is asserted.

register	command	expected	row reading
`exp:000000`	`python code/fk_truncation_moments.py`	exit 0	`refutes`; $k = 2$ vs $k = 8$: diff 1.9396, se
`exp:000001`	`python code/f_conj000000_delta_repro.py`	exit 0	`refutes`
`exp:000002`	`python code/fk_truncation_kdep_check.py`	exit 0	`inconclusive`
`exp:000003`	`python code/f_conj000000_delta_fixedsampler.py`	exit 0	`supports`
`exp:000004`	`python code/fk_truncation_corrected_check.py`	exit 0	`inconclusive`
`exp:000005`	same script	exit 0	`refutes`
`exp:000006`	`python scripts/conj000000_infeasible_record.py`	exit 0	`inconclusive`
`exp:000007`	same script	exit 1	`inconclusive`
`exp:000008`	same script	exit 0	`inconclusive`
`exp:000009`	`python code/conj000000_verdict_run.py`	exit 0	`inconclusive`
`exp:000010`	same script	exit 0	`inconclusive`

A.3 A sampling bias that refuted our first estimator. The first estimator sampled words by drawing exactly N letters uniformly from S and compared the resulting counts with the number of reduced words of length at most N. The two populations differ: uniform sampling of N-letter words concentrates on words of length exactly N before cancellation, whereas the target count is over all reduced words of length $\leq N$. The estimator was therefore rejected on its own output, and the rejection concerns our implementation, not any statement about $W^*(F_k)$. A corrected sampler produced internally consistent readings, which is a statement about the script only; internal consistency is not support for any conjecture.

A.4 Truncated-moment route. A second route summed Haar-random permutation matrices and looked at the spectrum of the real part. At $n = 64$ the standardised free cumulant of order four differed between $k = 2$ and $k = 8$ by approximately 1.94, with a pooled standard error of approximately 0.0344, i.e. a ratio of about 56.34. This separates k at fixed n, but the separation

is finite-n: the normalisation of the model contributes a factor $-1/(2k)$ to the cumulants, so a difference in k is an arithmetic consequence of the normalisation and not an invariant of the tuple. We therefore record the reading as a heuristic observation about the model, not as evidence for non-isomorphism.

A.5 Gate criteria and the last run. To decide whether a sequence of estimates could support any conclusion at all, four gate criteria were imposed on an estimator: (Q1) convergence, the relative standard error non-increasing in N with final value at most 0.1; (Q2) N-scaling consistency, the relative change of the estimate from $N = 32$ to $N = 64$ at most 0.2; (Q3) normalisation invariance, relative difference at most 0.05 between normalisation scales at fixed (k, N); (Q4) feasibility, at most one level reported as infeasible. The parameter range was $k \in \{2, 3, 4\}$, $m \in \{3, 4, 5\}$ with $m \neq k$, and $N \in \{8, 16, 32, 64\}$, with a fixed deterministic seed per level; 7 cells were measured. Readings with convergent error were obtained at $k = 2, N = 8$ (relative standard error 0.0348) and at $k = 3, N = 8$ and $k = 3, N = 16$ (0.0067 and 0.0308). All levels with $k = 4$ returned an estimate equal to 1.0 and carried no information. The cells $k = 2$ with $N = 16, 32, 64$ and $k = 3$ with $N = 32, 64$ were rejected by an analytic hit-rate precheck as infeasible. The deterministic gate therefore returned the verdict 'insufficient data' with the gate disqualified, the failing reason being that no estimator passed Q1 together with Q2 and Q3; the obstruction is the absence of a qualified estimator, not a shortage of data points.

A.6 What would change the picture. An estimator satisfying (i)–(iv) of Open problem 4.1 would allow the value k to be checked at finite level; a positive check would not prove Conjecture 3.11, but a failure of stability in N at every feasible level would show that the microstates route cannot be pushed numerically at accessible sizes.